UPDATING FORTIGATE FIREWALL

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Lab Goal

The goal of this lab was to successfully upgrade the FortiGate 40F to any version in the 7.x.x without any loss of configuration settings through the upgrade path.

Background

The basic setup of Fortigate firewalls is much simpler compared to other firewall brands like Palo Alto. To connect to the firewall, an ethernet cord needs to be directly connected to the PC, then entering the GUI through the default Fortinet firewall IP 192.168.1.99. The only other step on connecting the firewall to the internet is to connect an ethernet cable between the WAN port and a port on a router that can access the internet. Then, we can configure the WAN interface to automatically receive an IP address via DHCP.

Generally, firewalls are used to protect networks from outside attacks, such as filtering traffic to negate the effects of DDOS attacks, a malicious entity attempting to remotely log into a device on your network, or traffic based on an unknown IP address that can contain phishing attacks. A firewall is a must-have for every corporation that aims to maintain the security of their networks. After all, any connection to the public internet poses a security risk in which outside attackers have access to your network.

However, not every firewall has the same functionalities and benefits when compared to others. For instance, every major competitive firewall brand has different selling points for businesses. Juniper firewalls use AI to predict threats and aims to stop those threats before they can affect the networks and Palo Alto firewalls can classify traffic with different tags and then filter the traffic based on those tags. Fortigate firewalls have a strong point in their

By default, our Fortinet firewall's version was 6.4.6. This version is very outdated and poses a security risk, especially because 6.4 engineering support ended in 2022, which prepares for the end of support in September 2024. It is critical to ensure that the firewall is off the 6.4 branch because even if all major security bugs are patched, its still important to ensure that minor updates that can patch possible security breaches may not be pushed out. Thus, its important to upgrade the firewall into the 7.x.x version to prepare for the end of life in 6.4.

While in most cases, its optimal to update devices to the latest version which often provide the latest updates and thus the most secure firmware, as of 2/26/24, there are numerous bugs out on the latest version which proves to be impractical to upgrade to and even poses a critical security risk. (19 bugs in the 6000-7000 platforms)

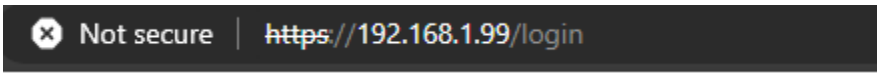
Upgrading a firewall is similar to upgrading a computer's operating system. You cannot directly upgrade from Windows 8.0 to Windows 10. You must follow an upgrade path by upgrading it from Windows 8.0 to Windows 8.1 first or you may lose configuration Firewalls follow the same logic; a 6.4.6 firewall cannot directly upgrade to 7.2.8M. A firewall on 6.4.6 must first upgrade to 6.4.8, 6.4.10, 6.4.12M, 6.4.14M, 7.0.0, 7.0.10M, then 7.2.5.

Lab Commands

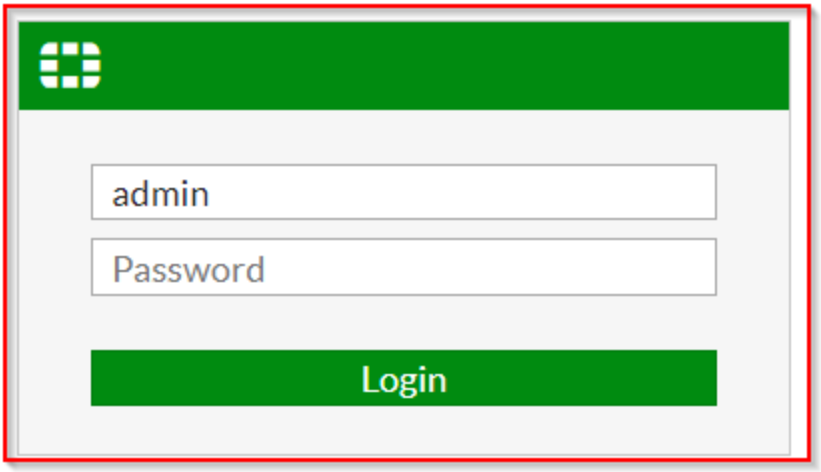
Basic Setup:

Plug the fortinet firewall into the power outlet and connect the WAN port to a router that will route its traffic to the internet. Then plug the other ethernet port into your host PC.

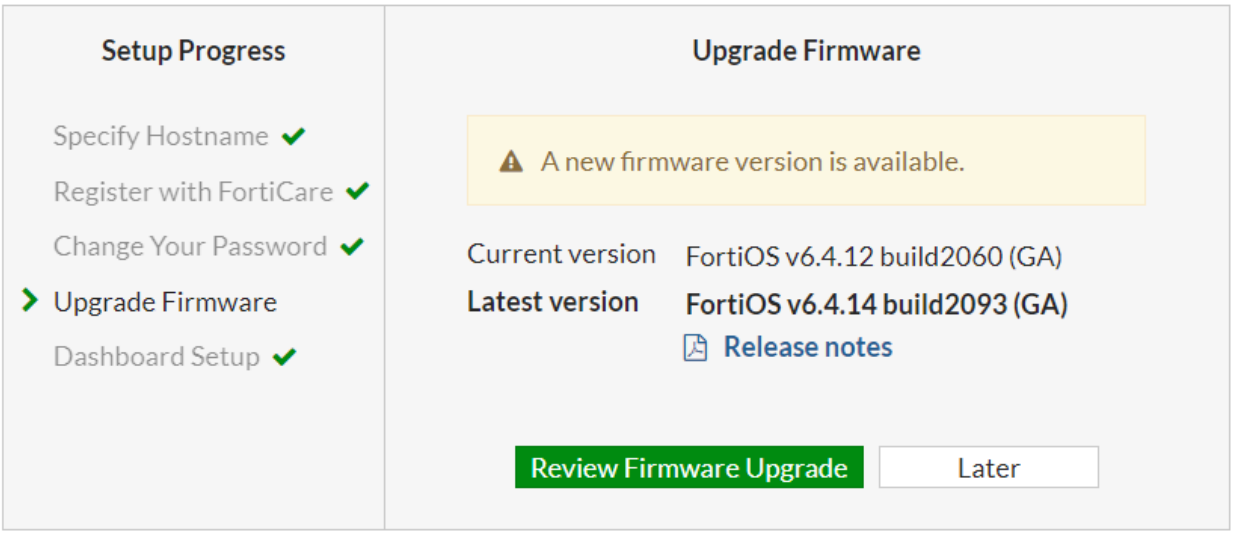
Type https://192.168.1.99 into a web browser. This is the default ip of the Fortigate Gui.



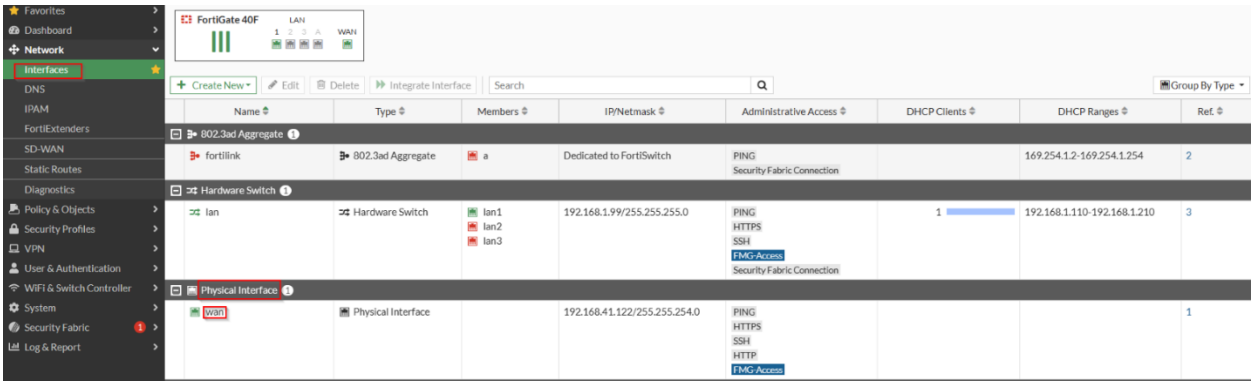
Type in the default username "admin" with no password and then log in.



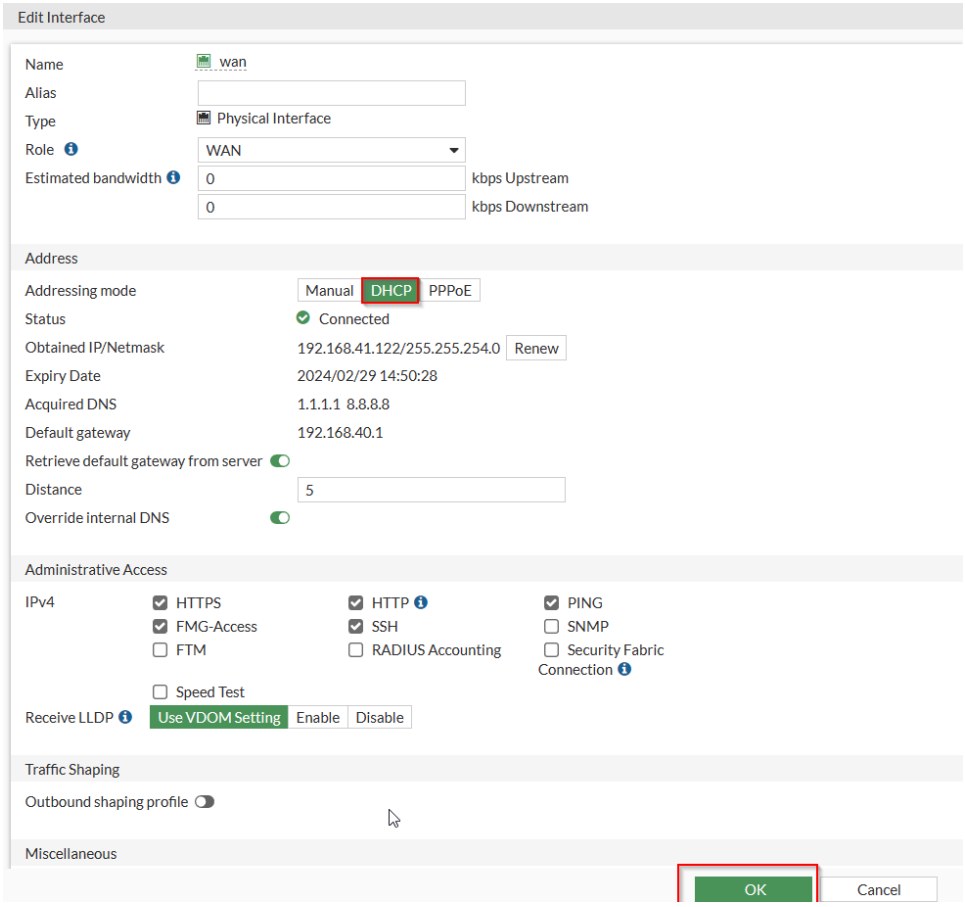
Create a hostname, register with Forticare, and change your password because the default password is empty. Notice the outdated firmware.



Enter Interfaces > Physical Interfaces > WAN



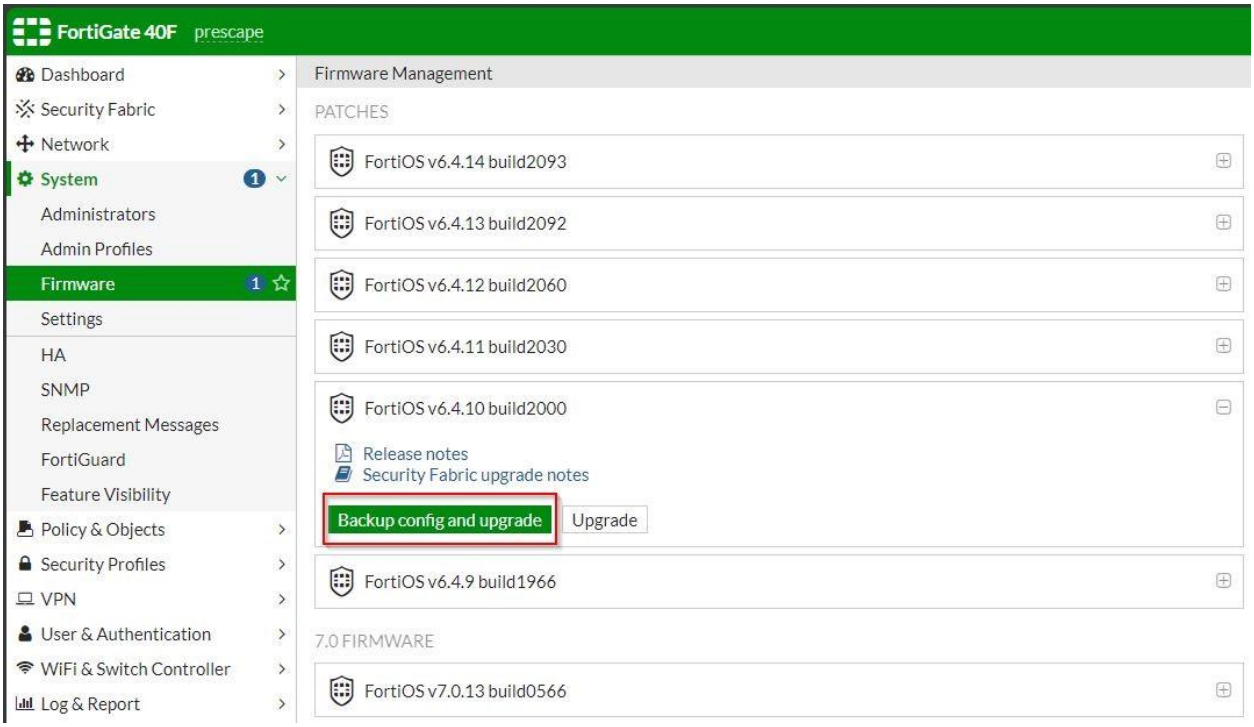
Change the Addressing mode from Manual to DHCP. Allow HTTPS, HTTP, and PING. Then press OK.



4.

Next, we had to update Fortigate OS through this upgrade path to ensure no configuration data is lost during the updates. All of these next steps will take place in the System > firmware section.

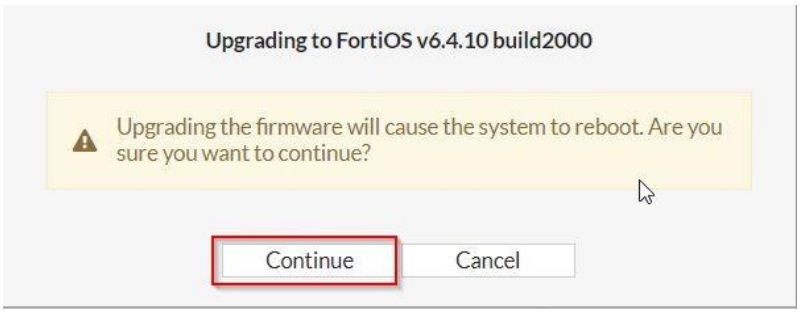
6.4.6
Build 1879
6.4.8
Build 1914
6.4.10M
Build 2000
6.4.12M
Build 2060
6.4.14M
Build 2093



Update FortiGate OS from 6.4.8 to 6.4.10

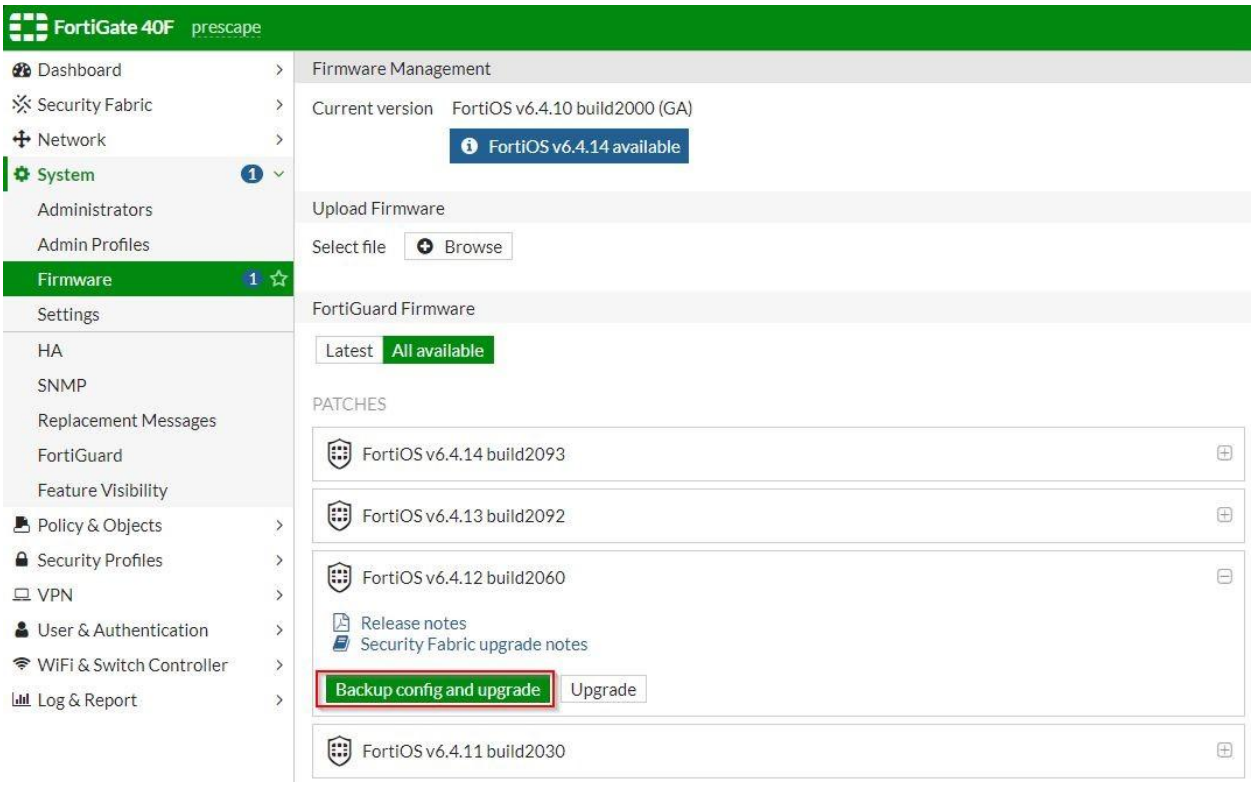
5.

When prompted with the confirmation message, press continue.



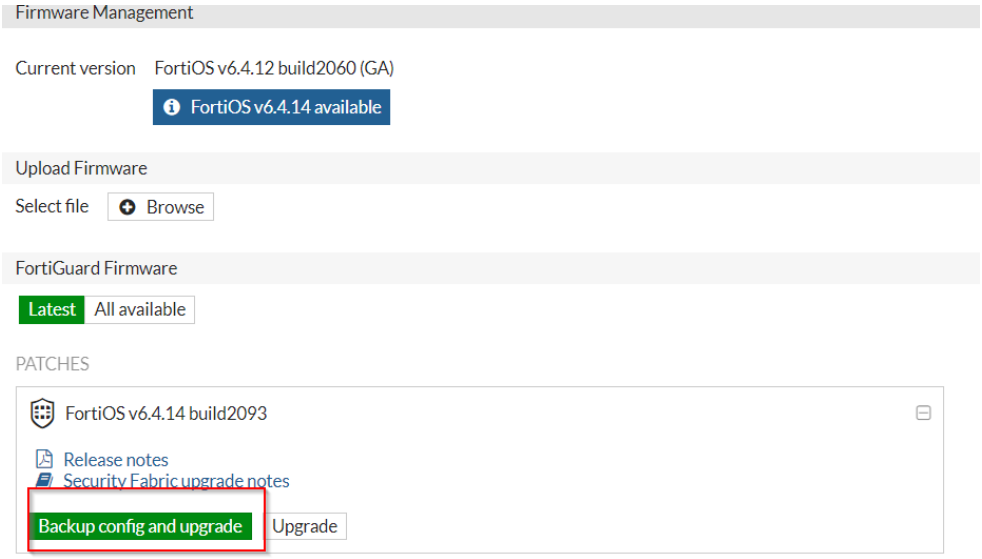
6.

Once the FortiGate has finished restarting, update the 6.4.10 build to 6.4.12



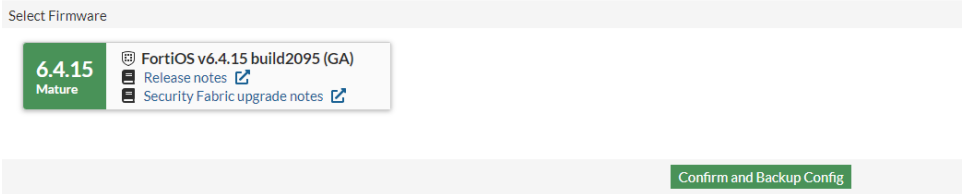
7.

Upgrade the firewall from 6.4.12 to 6.4.14. Backup the config and then upgrade.



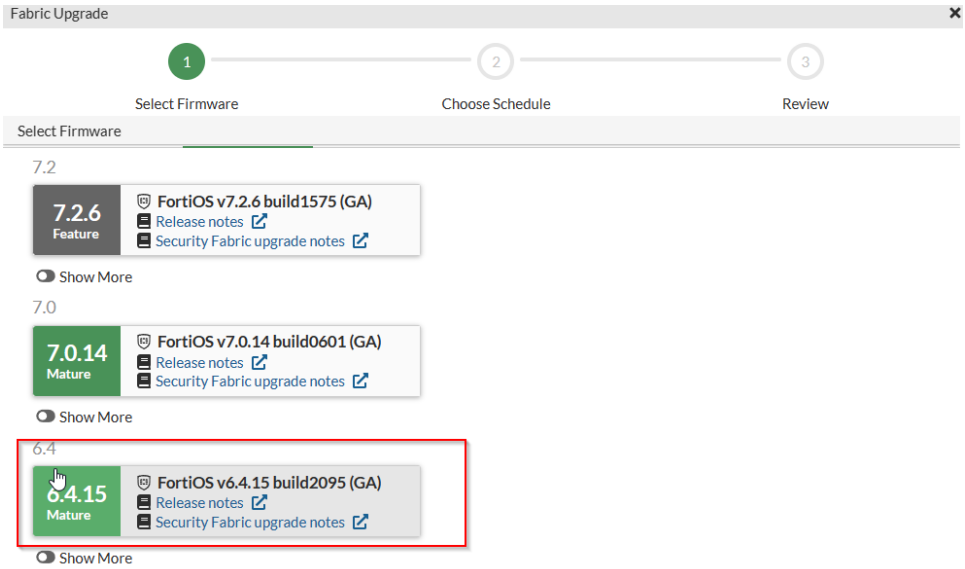
8.

Upgrade the firmware from 6.4.14 to 6.4.15. Backup the config then upgrade.



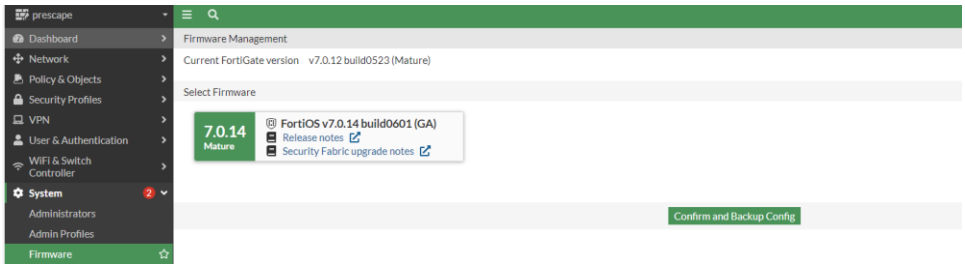
9.

Upgrade from 6.4.15 to 7.0.0. Backup the config, then save.



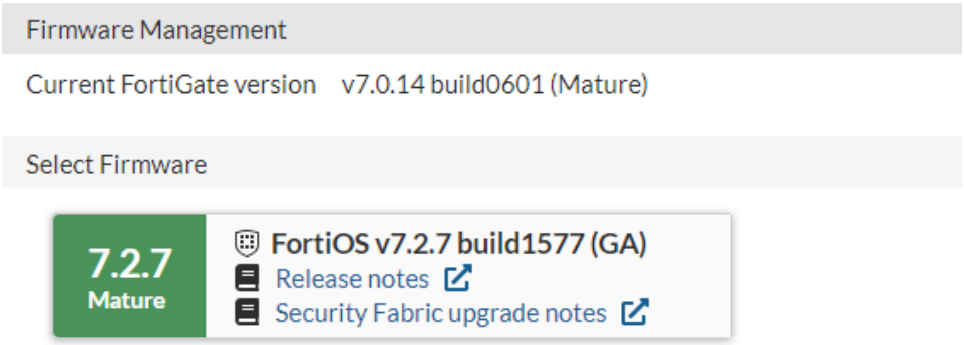
10.

Upgrade from 7.0.0 to 7.0.14. Backup the config and then save.

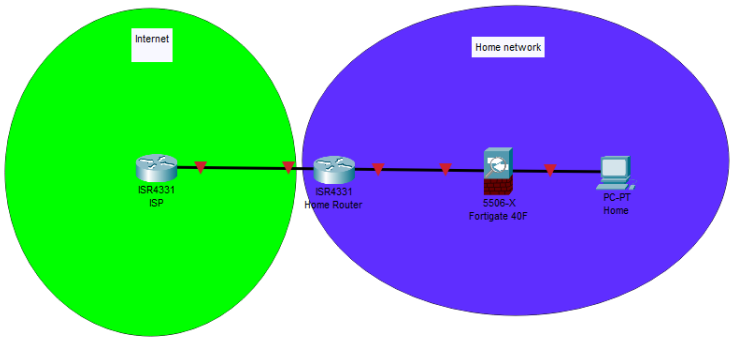


11.

Upgrade from 7.0.14 to 7.2.7. Backup the config then save.



Topology



Problems

Fortigate GUI would not let us in without a fortinet account

created an account on fortinet and linked it to the fortigate firewall

Conclusion

Firewalls are a mandatory device in any network that contains any devices that need to be secured on it. Fortigate firewalls have a reputation for being the most reliable, especially since it is the most used firewall in the world. Fortigate may be desirable because of its Fortiguard feature, using AI to detect millions of possible threats, but a fortigate firewall still cannot function well if its firmware is out of date. Firmware version is the most important aspect of ensuring a secure and reliable firewall stays that way. If firmware is outdated, then newly added features or threat detection services will not be available, which poses a security risk. Thus, it is paramount to make sure that all devices are in a safe version not only of possible end of support, but the immense security risks of staying on outdated versions.

Fortigate Update to 7.x.x

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Period 5

Mr. Mason